E.G.S. PILLAY ENGINEERING COLLEGE
***(Autonomous), NAGAPATTINAM* – 611 002.**
(Affiliated to Anna University, Chennai | Accredited by NAAC with „A++" Grade
Accredited by NBA | Approved by AICTE, New Delhi)

REGULATIONS - R2023
B.E – ELECTRICAL & ELECTRONICS ENGINEERING
SIXTH SEMESTER CURRICULUM

SEMESTER VI												
Course Code		**Course Name**	**L**	**T**	**P**	**H**	**C**	**Maximum Marks**			**Category**
								CA	**ES**	**Total**	
Theory Course											
1	**2302EE601**	Solid State Drives	3	0	0	3	**3**	40	60	100	PCC
2	**2302EE602**	Protection and Switchgear	3	2	0	5	**4**	40	60	100	PCC
3	-	**Elective –III**	3	0	0	3	**3**	40	60	100	PEC
4	-	**Elective –IV**	3	0	0	3	**3**	40	60	100	PEC
5	-	Open Elective –II -	3	0	0	3	**3**	40	60	100	OEC
6	**2301MC613**	Mandatory course-II - Mental Health & Emotional Wellbeing	2	0	0	2	**0**	100	0	100	MC
Theory Cum Laboratory Course											
7	**2302EE603**	Embedded Systems	3	0	2	5	**4**	50	50	100	PCC
Practical Course											
8	**2302EE651**	Power Systems Laboratory	0	0	2	2	**1**	60	40	100	PCC
9	**2304EE652**	Mini-Project	0	0	2	2	**1**	50	50	100	EEC
10	**2304GE601**	Professional Development – 4	0	0	2	2	**1**	100	0	100	EEC
11	**2301LS601**	Life Skill : 06	0	0	0	0	**0**	100	0	100	-
		Total	20	02	08	30	**23**	660	440	1100	

2302EE601	SOLID STATE DRIVES	L	T	P	C
		3	0	0	3

PREREQUISITE:

	Power Electronics
	Electrical Machinery – I & II

COURSE OBJECTIVES:

1	To understand the fundamentals of motor load system
2	To explain about power converters fed DC and AC drives
3	To design a controllers for closed loop operation of DC and AC drives

COURSE OUTCOMES:

On the successful completion of the course, students will be able to

CO1	Understand the drive system components, torque-speed dynamics, and stability.
CO2	Explain about single and multi-quadrant operation of power converter fed dc drives.
CO3	Simulate and model drive systems using software tools for practical energy system applications.
CO4	Explain scalar and vector control techniques for induction motor drives.
CO5	Evaluate the performance of synchronous and special motor drives in real-world applications.

COs Vs. POs MAPPING:

COs	PO1	PO2	PO3	PO4	PO5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11
CO1	3	3	-	-	1	2	-	-	-	-	-
CO2	3	3	1	2	1	2	-	-	-	-	-
CO3	3	3	2	2	3	-	-	-	-	2	2
CO4	3	3	1	-	-	-	-	-	-	-	-
CO5	3	3	-	-	-	2	-	-	-	2	-

COs Vs. PSOs MAPPING:

COs	PSO1	PSO2
CO1	3	-
CO2	3	-
CO3	3	3
CO4	3	-
CO5	3	-

MODULE I	**DRIVE CHARACTERISTICS**	**08 Hours**

Components of an electric drive system. Dynamics of motor-load system: torque-speed characteristics, steady-state and transient stability. Duty cycles, load equalization, thermal considerations. Multi-quadrant operation and braking methods. Drive design overview: applications in EVs, robotics, automation. Selection of motor rating.

MODULE II	**DC MOTOR DRIVE**	**09 Hours**

Converter fed drive- Review of one and two quadrant converter and its characteristics; Steady state analysis of single phase and three phase converter fed separately excited dc motor drive - Continuous and discontinuous

conduction, four quadrant operations of converter; Armature control, field control and regenerative braking in DC motors using phase angle control. **Chopper fed drive-** Review of dc chopper and its control strategies; Motoring mode, braking mode and four quadrant operation of chopper fed drive.		
MODULE III	**CLOSED LOOP CONTROL OF DC DRIVE**	**08 Hours**
Transfer functions for converters, motors, and drive systems. Cascaded speed and current control loops. Design of PI controllers, field weakening techniques. Use of simulation tools (e.g., MATLAB, PLECS) for dynamic modeling. Case studies: EV drive cycle analysis, regenerative energy recovery		
MODULE IV	**INDUCTION MOTOR DRIVE**	**12 Hours**
Review of induction motor equivalent circuit and torque-speed characteristics. Scalar control: stator voltage control, V/f control. VSI and cycloconverter-fed drives. Rotor resistance control, slip power recovery. Introduction to vector control and field-oriented control (FOC) principles		
MODULE V	**SYNCHRONOUS AND SPECIAL MOTOR DRIVE**	**08 Hours**
V/f and self-control methods for synchronous motor drives. Margin angle and power factor control. Permanent Magnet Synchronous Motor (PMSM) and BLDC drives. Traction, robotics, and precision motion control applications. Regenerative braking in PMSM and SRM.		
	TOTAL:	**45 HOURS**

REFERENCES:

1. G.K Dubey, "Fundamentals of Electrical Drives", 2nd Edition, Narosa Book Distributors, 2013.
2. N. K. De, P. K. Sen, "Electric Drives", 16th Edition, PHI Learning Pvt. Ltd., 2014.
3. R. Krishnan, "Electric Motor Drives: Modeling, Analysis and Control", Pearson Education, 2015.
4. Bimal K. Bose — Power Electronics and Motor Drives: Advances and Trends (2nd Edition, 2020)
Comprehensive, modern treatment of power converters, motor drives, control methods, and applications; includes fuzzy logic, neural network control, and industrial case studies.
5. P.C Sen "Thyristor DC Drives", John wiely & Sons, New York, 1981.
6. R. Krishnan, "Permanent Magnet Synchronous and Brushless DC motor Drives", CRC Press, NewYork, 2010.
7. https://nptel.ac.in/courses/108/104/108104140/
8. M.H. Rashid, Power Electronics Handbook, Elsevier, 2017.
9. Cambridge / Georgia Tech Open Course Materials on Electric Drives.

2302EE602	PROTECTION AND SWITCHGEAR	L	T	P	C
		3	2	0	4

PREREQUISITE:

Power System Analysis

COURSE OBJECTIVES:

1	To provide the basic principles of arc interruption, circuit breaking principles, operation of various types of circuit breakers.
2	To study the classification, operation, construction and application of different types of relays
3	To explain various types of fault occurs in power components and different types of protective schemes.

COURSE OUTCOME:

CO1	Explain the various protection schemes involves in the power system
CO2	Classify various electromagnetic relays working principle their operating characteristics
CO3	Differentiate various static relay and microprocessor based relays used for modern power System.
CO4	Analyze the arc phenomenon process and compare the working principles of various circuit breakers
CO5	Demonstrate the power protection schemes incorporates various power system components

COs Vs. POs MAPPING:

COs	PO1	PO2	PO3	PO4	PO5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11
CO1	3	2	-	-	-	1	-	-	-	-	1
CO2	3	2	-	-	-	-	-	-	-	-	1
CO3	2	3	2	-	-	-	-	-	-	-	-
CO4	3	3	2	2	-	-	-	-	-	-	1
CO5	3	3	3	-	-	-	-	-	-	-	-

COs Vs. PSOs MAPPING:

COs	PSO1	PSO2
CO1	2	-
CO2	3	-
CO3	3	-
CO4	3	-
CO5	2	-

MODULE I	**PROTECTION AGAINST OVER VOLTAGE AND GROUNDING METHODS**	**12 Hours**

Generation of over voltages in power systems, Protection against lightning over voltages, Lightning arresters, Insulation coordination, Grounded and ungrounded neutral systems, Methods of neutral grounding: Solid–resistance–Reactance, Arcing grounds and grounding Practices.

MODULE II	**ELECTROMAGNETIC RELAYS**	**12 Hours**

Relay phenomenon, types of electromagnetic relays-attracted armature type, balanced beam type, induction type relays, over current relays, Directional relays, differential relays, distance relays, Negative sequence relay, under frequency relay Universal torque equation,

MODULE III	**STATIC RELAYS AND DIGITAL PROTECTION**	**12Hours**

Static over current relay– Static distance relay Introduction to digital relays: Basic Layout and elements of digital relays- Numerical relay- Microprocessor based digital relays for transmission line protection,, over current relay protection for micro grids, travelling wave based digital protection		
MODULE IV	**CIRCUIT BREAKERS**	**12 Hours**
Circuit Breakers: MCB, ELCB, RLCB, arc interruption, Restriking Voltage and Recovery voltages, Average and Max. RRRV- Current chopping and Resistance switching- oil circuit breakers, Air Blast, Vacuum and SF6 circuit breakers		
MODULE V	**PROTECTION OF POWER APPARATUS**	**12 Hours**
Generator Protection: Protection of generators against stator faults - Rotor faults and abnormal conditions- restricted earth fault and inter turn fault protection- Numerical examples. Transformer Protection of transformers- Percentage differential protection- CT's ratio- Buchholz relay protection- Feeder and Bus bar Protection		
REFERENCES:		

1. B. Rabindranath, M. Chander, "Power system protection and switchgear," New Age International (P) Ltd publishers, 2005 edition.
2. V. K. Mehta, Rohit Mehta, "Principles of power systems", S. Chand &co , New Delhi 2016 edition .
3.Lakneshwar Prakash Singh, " Digital Protection Protective Relaying From Electromechanical To Microprocessor," New Age International (P) Limited,2006 Reprint edition
4.Digital protection and signaling " Power system protection ", The institution of electrical engineers, volume London, United kingdom, 2005 Reprint edition.
5. Prof. Bhaveshkumar R. Bhalja, "Power System Protection and Switchgear," IIT Roorkee, NPTEL video lectures.

2302EE603	EMBEDDED SYSTEMS	L	T	P	C
		3	0	2	4

PREREQUISITE:

	Microprocessors and Microcontrollers

COURSE OBJECTIVES:

1	To introduce the Building Blocks of Embedded System
2	To Introduce Bus Communication in processors, Input/output interfacing
3	To design innovative applications by interfacing the processors with real world

COURSE OUTCOMES:

On the successful completion of the course, students will be able to

CO1	Explain the architecture, features and operating principles of embedded systems, including processor selection, memory hierarchy, timers, watchdogs, and development tools.
CO2	Apply embedded networking concepts to interface I/O ports, handle multiple interrupts, and implement communication using RS232, RS485, USB, I²C, CAN, Wi- Fi, Bluetooth, and ZigBee protocols.
CO3	Make Use of appropriate interfacing techniques to design ARM-based embedded systems involving analog and digital peripherals..
CO4	Apply peripheral interfacing techniques of the LPC2148 peripherals for real-world embedded applications.
CO5	Develop embedded applications using ARM-based Embedded C programming for GPIO, UART, timers, and ADC

COs vs. POs MAPPING:

COs	PO1	PO2	PO3	PO4	PO5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11
CO1	3	2	1	-	-	-	-	-	-	-	-
CO2	3	2	1	-	-	-	-	-	-	-	-
CO3	3	2	1	-	-	-	-	-	-	-	-
CO4	3	2	1	-	2	-	-	-	-	-	-
CO5	3	2	1	-	3	-	-	-	-	-	-

COs Vs PSOs MAPPING:

COs	PSO1	PSO2
CO1	-	2
CO2	-	3
CO3	-	3
CO4	-	3
CO5	-	3

COURSE CONTENTS:

MODULE I	INTRODUCTION TO EMBEDDED SYSTEMS	9 Hours

Introduction to Embedded Systems –built in features for embedded Target Architecture - selection of Embedded processor – DMA- memory devices – Memory management methods-memory mapping, cache replacement policies- Timer and Counting devices, Watchdog Timer, Real Time Clock Software Development tools-IDE, assembler, compiler, linker, simulator, debugger, In circuit emulator, Target Hardware Debugging.

MODULE II	**EMBEDDED NETWORKING BY PROCESSORS**	**9 Hours**
Embedded Networking: Introduction, I/O Device Ports & Buses- multiple interrupts and interrupt service mechanism – Serial Bus communication protocols -RS232 standard–RS485–USB–Inter Integrated Circuits (I2C)- CAN Bus –Wireless protocol based on Wifi , Bluetooth, Zigbee – Introduction to Device Drivers.		
MODULE III	**ARM PROCESSOR**	**9 Hours**
Architecture of ARM Controller – Registers, Pipeline organization 3 stage & 5 stage, Thumb mode of operation - D/A and A/D converter, sensors, actuators and their interfacing .		
MODULE IV	**REAL WORLD INTERFACING USING ARM PROCESSOR**	**9 Hours**
Interfacing the peripherals to LPC2148: GSM and GPS using UART, on-chip ADC using interrupt (VIC), EEPROM using I2C, SD card interface using SPI, on-chip DAC for waveform generation.		
MODULE V	**EMBEDDED PROGRAMMING**	**9 Hours**
ARM Peripherals programming in Embedded C-GPIO-Serial Communication-Timer-ADC. Case study- Digital clock, Temperature sensing, Light sensing,		
		45 HOURS
LAB SYLLABUS		
1. AUTOMATED RGB LED BLINKING USING ETS IOT KIT		
2. MONITORING TEMPERATURE & HUMIDITY OF ENVIRONMENT USING SENSOR		
3. MONITORING DISTANCE OF AN OBJECT USING ULTRASONIC SENSOR		
4. RAIN SENSOR INTERFACING WITH ETS IOT KIT		
5. INTERFACING OLED DISPLAY WITH ETS IOT KIT		
		15 HOURS
		TOTAL : 45+15 = 60 HOURS

REFERENCES:

1. Raj Kamal, Embedded Systems Architecture, Programming, and Design. (2/e), Tata McGraw Hill, 2008.
2. K.V. Shibu, Introduction To Embedded Systems, Tata McGraw, 2009.
3. Peter Barry and Patric Crowley, Intel architecture for Embedded system .
4. Tammy Noergaard "Embedded Systems Architecture: A Comprehensive Guide for Engineers and Programmers", Elsevier (Singapore) Pvt. Ltd. Publications, 2005.
5. Jonathan W. Valvano, "Embedded Microcomputer Systems Real Time Interfacing", 3 rd Edition, Cengage Learning, 2012
6. Peter Marwedel, "Embedded System Design", Science Publishers, 2007.
7. http://www.tomshardware.com/reviews: Pierre Dandumont, Intel and Declining Power Consumption, 2008.
8. Lori Matassa and Max Domeika, Break Away with Intel® Atom™ Processors,2010, Intel press.

2303EE003	DESIGN OF MOTOR AND POWER CONVERTERS FOR ELECTRIC VEHICLE	L	T	P	C
		3	0	0	3

PREREQUISITE:

1. Power electronics
2. Special Electrical Machines

COURSE OBJECTIVES:

1	To analyze Electric Vehicle systems including powertrain architecture, vehicle dynamics and operating requirements for motor and converter design.
2	To design and evaluate electric motors used in Electric Vehicle applications based on performance characteristics, loading conditions and design constraints.
3	To Provide design-oriented knowledge of power electronic converters used in Electric Vehicles

COURSE OUTCOMES:

On the successful completion of the course, students will be able to

CO1	Scrutinize Electric Vehicle powertrain requirements and vehicle dynamics to determine motor specifications.
CO2	Evaluate and compare electric motors suitable for EV applications based on torque–speed characteristics, efficiency and operating regions.
CO3	Design basic electric motor parameters for Electric Vehicle applications using output equations and performance criteria.
CO4	Examine the basic power electronic converters (AC–DC, DC–DC and DC–AC) used in Electric Vehicle systems.
CO5	Develop power converter configurations for Electric Vehicle applications considering performance and application requirements.

COs Vs. POs MAPPING:

COs	PO1	PO2	PO3	PO4	PO5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11
CO1	3	2	1	-	-	-	-	-	-	-	-
CO2	3	2	1	-	-	-	-	-	-	-	-
CO3	3	3	2	1	-	-	-	-	-	-	-
CO4	3	3	1	-	-	-	-	-	-	-	-
CO5	3	2	2	1	-	-	-	-	-	-	-

COs Vs. PSOs MAPPING:

COs	PSO1	PSO2
CO1	3	-
CO2	3	-
CO3	3	-
CO4	-	3
CO5	-	3

COURSE CONTENTS:		
MODULE I	**INTRODUCTION**	**9 Hours**
Overview of Electric Vehicles (EV), EV powertrain architecture , Role of electric motors in EV propulsion systems, Standard drive cycles- Dynamics of Electric Vehicles-Tractive force, performance and characteristics, Selection criteria of motors for EV applications		
MODULE II	**MOTORS FOR ELECTRIC VEHICLES**	**9 Hours**
Introduction – Induction Motor, Permanent Magnet Synchronous Motors (PMSM), and Brushless DC Motors; Speed and Torque control of above and below rated speed-Speed control of EV in the constant power region of electric motors; Choice of electric machines for EVs.		
MODULE III	**DESIGN OF MOTORS FOR EV APPLICATIONS**	**9 Hours**
Induction Motors: Output equation, Choice of specific loadings, main dimensions of three phase induction motor, Stator winding design, Design of rotor slots Permanent Magnet Synchronous Motor: Output equation, Choice of specific loadings, short circuit ratio, design of main dimensions, Armature design.		
MODULEIV	**INTRODUCTION TO POWER CONVERTERS FOR ELECTRIC VEHICLES**	**9 Hours**
Role of power converters in EV systems, Overview of EV power electronic architecture, AC–DC converters, DC–DC converters, DC–AC inverters, On-board and off-board converters, Losses in power converters		
MODULE V	**DESIGN OF POWER CONVERTERS FOR ELECTRIC VEHICLES**	**9 Hours**
Design of DC–DC converters for EVs, Buck, Boost, Buck-Boost converters, Bidirectional DC–DC converters, Inverter design for EV motor drives, thermal management and protection of power electronic devices.		
		TOTAL: 45 HOURS

REFERENCES:

1. C. C. Chan, K. T. Chau, Modern Electric Vehicle Technology, Oxford University Press, New York, 2001.
2. Mehrdad Ehsani, Yimin Gao, Sebastien E. Gay, Ali Emadi, Modern Electric, Hybrid Electric and Fuel Cell Vehicles – Fundamentals, Theory and Design, CRC Press, 2nd Edition, 2010.
3. P. S. Bimbhra, Electrical Machinery, Khanna Publishers, New Delhi, 7th Edition, 2011.
4. K. Sawhney, A Course in Electrical Machine Design, Dhanpat Rai & Co., New Delhi, 6th Edition, 2017.
5. Muhammad H. Rashid, Power Electronics – Circuits, Devices and Applications, Pearson Education, 4th Edition, 2014.
6. Robert W. Erickson, Dragan Maksimović, Fundamentals of Power Electronics, Springer (formerly Kluwer Academic Publishers), 2nd Edition, 2001.
7. Ned Mohan, Tore M. Undeland, William P. Robbins, Power Electronics: Converters, Applications and Design, John Wiley & Sons, 3rd Edition, 2003.
8. Robert Bosch GmbH, Automotive Handbook, Wiley Publications, 9th Edition, 2014.
9. G. K. Dubey, Fundamentals of Electrical Drives, Narosa Publishing House, New Delhi, 2nd Edition, 2009.
10. *https://nptel.ac.in/courses/108106182/ – Electric Vehicles and Renewable Energy, IIT Madras*
11. *https://nptel.ac.in/courses/108102145/ – Power Electronics, IIT Delhi*

2303EE004	**DESIGN OF ELECTRIC VEHICLE CHARGING SYSTEMS**	**L**	**T**	**P**	**C**
		3	**0**	**0**	**3**

PREREQUISITE:

1	Electric Vehicle Fundamentals
2	Power Electronics

COURSE OBJECTIVES:

1	To understand about charging stations and communication standards
2	To learn the concepts related to wireless charging systems
3	To know about power electronics in EV charging

COURSE OUTCOMES:

On the successful completion of the course, students will be able to

CO1	Explain the charging stations and standards
CO2	Discuss the concepts of charging using renewable energy storage systems.
CO3	Explain the concepts related to wireless charging systems
CO4	Discuss about communication**,** standards & protocols
CO5	Perform power factor correction in charging systems

COs Vs POs MAPPING:

COs	**PO1**	**PO2**	**PO3**	**PO4**	**PO5**	**PO 6**	**PO 7**	**PO 8**	**PO 9**	**PO 10**	**PO 11**
CO1	3	2	1	-	-	-	-	-	-	-	-
CO2	3	3	2	-	-	-	-	-	-	-	-
CO3	3	2	2	-	2	-	-	-	-	-	-
CO4	3	2	1	-	3	-	-	-	-	-	-
CO5	3	3	3	-	3	-	-	-	-	-	-

COs Vs PSOs MAPPING:

COs	**PSO1**	**PSO2**
CO1	2	1
CO2	3	2
CO3	1	2
CO4	1	3
CO5	2	3

COURSE CONTENTS:

MODULE I	**CHARGING STATIONS AND STANDARDS**	**9 Hours**

Introduction-Charging technologies- Conductive charging, EV charging infrastructure,

International standards and regulations - Inductive charging, need for inductive charging of EV, Modes and operating principle, Static and dynamic charging, Bidirectional power flow, Types of commercial chargers, International standards and regulations.		
MODULE II	**CHARGING USING RENEWABLE AND STORAGE SYSTEMS**	**9 Hours**
Introduction- - EV charger topologies, EV charging/discharging strategies - Integration of EV charging-home solar PV system (HSP), Operation modes of EVC-HSP system, Control strategy of EVC-HSP system - fast-charging infrastructure with solar PV and energy storage.		
MODULE III	**WIRELESS CHARGING**	**9 Hours**
Introduction - Inductive, Magnetic Resonance, Capacitive types. Wireless Chargers for Electric Vehicles -Battery Technology in WPT -Charging Modes in WPT - Benefits of WPT. - WPT Operation Modes - Standards for EV Wireless Chargers, SAE J2954, IEC61980.ISO19363.		
MODULE IV	**COMMUNICATION, STANDARDS & PROTOCOLS**	**9 Hours**
Open Charge Point Protocol (OCPP)-Familiarizing with the OCPP protocol for communication between charging stations and the cloud. **Vehicle to Charger Communication**-Understanding the communication protocols used between the vehicle and the charging station. CAN Communication-Learning about CAN communication for automotive applications and its role in EV charging. **Embedded Processors in EVs-** Exploring the use of microcontrollers and embedded systems in EV charging systems.		
MODULE V	**POWER ELECTRONICS IN EV CHARGING**	**9 Hours**
single-line diagrams for EV charging systems, exploring different DC-DC converter topologies used in EV chargers (e.g., LLC, DAB) and their control strategies. Understanding the sources of EMI/EMC in charging systems and designing appropriate filters. Need for power factor correction- Boost Converter for Power Factor Correction, Sizing the Boost Inductor, Average Currents in the Rectifier and calculation of power losses.		
		TOTAL: 45 HOURS

REFERENCES:

1. **James Larminie**, *Electric Vehicle Technology Explained,* Second edition, Wiley & sons, 2012.
2. Wie Liu, “Hybrid Electric Vehicle System Modeling and Control”, Second Edition, John Wiley & Sons, 2017.
3. Mobile Electric Vehicles Online Charging and Discharging, Miao Wang Ran Zhang Xuemin (Sherman) Shen, Springer, 1st Edition, 2016.
4. Alicia Triviño-Cabrera, José M. González-González, José A. Aguado, Wireless Power Transferor Electric Vehicles: Foundations and Design Approach, Springer Publisher, 1st Edition. 2020.
5. https://nptel.ac.in/courses/108107715
6. https://onlinecourses.nptel.ac.in/noc21_ee112/preview

2303EE010	DISTRIBUTED GENERATION SYSTEMS	L	T	P	C
		3	0	0	3

PREREQUISITE:

1. Generation ,Transmission and distribution

COURSE OBJECTIVES:

1	To familiarize with the concept of Distributed Generation
2	To expose the various distributed energy resources
3	To focus on the planning and protection of Distributed Generation

COURSE OUTCOME:

CO1	Explain the basic concepts of electric power generation and distribution systems.
CO2	Describe the various Distributed Generation (DG) resources and their working principles.
CO3	Discuss the planning considerations and requirements of Distributed Generation systems.
CO4	Explain the principles of protection and operation of Distributed Generation in power systems.
CO5	Summarize the impacts of integrating Distributed Generation with the utility grid.

COs Vs POs MAPPING:

COs	PO1	PO2	PO3	PO4	PO5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11
CO1	3	2	-	-	-	1	2	-	-	-	1
CO2	3	2	-	-	-	-	2	-	-	-	-
CO3	3	3	2	2	-	-	1	-	-	-	2
CO4	3	3	2	2	-	-	-	-	-	-	1
CO5	3	2	2	2	-	1	3	-	-	-	2

COs Vs PSOs MAPPING:

COs	PSO1	PSO2
CO1	2	-
CO2	3	-
CO3	3	-
CO4	3	-
CO5	3	-

MODULE I	INTRODUCTION TO DISTRIBUTED GENERATION	9 Hours

DG definition - Reasons for distributed generation-Benefits of integration - Distributed generation and the distribution system - Technical, Environmental and Economic impacts of distributed generation on the distribution system - Impact of distributed generation on the transmission system-Impact of distributed generation on central generation.

MODULE II	DISTRIBUTED ENERGY RESOURCES	9 Hours

Combined heat and power (CHP) systems-Wind energy conversion systems (WECS)- Solar photovoltaic (PV) systems-Small-scale hydroelectric power generation-Other renewable energy sources-Storage devices-Inverter interfaces.

MODULE III	**DG PLANNING**	**9 Hours**
Generation capacity adequacy in conventional thermal generation systems-Impact of distributed generation-Impact of distributed generation on network design, Planning criteria for DG integration, Load flow analysis in distribution systems with DG, Impact of DG on voltage profile, losses, and reliability, Power quality issues related to DG.		
MODULE IV	**DG PROTECTION**	**9 Hours**
Protection coordination in presence of DG, Introduction to DG Protection, Fault Characteristics in DG Systems, Protection Coordination with DG, Anti-Islanding Protection, Protection Schemes for Different DG Types, Case Studies and Practical Examples.		
MODULE V	**IMPACTS OF GRID INTEGRATION**	**9 Hours**
Overview of Grid Integration, Impact on Power Quality, Voltage Regulation and Stability Issues, Effects on System Protection, Impact on Load Flow and Losses, Influence on Grid Reliability and Resilience, Environmental and Economic Impacts.		
		TOTAL: 45 HOURS

FURTHER READING / CONTENT BEYOND SYLLABUS / SEMINAR :

	Role of micro grid in power market competition
	Case study: Techniques used for generation and distribution of power.

REFERENCES:

1. Nick Jenkins, JanakaEkanayake, GoranStrbac, “Distributed Generation”, Institution of Engineering and Technology, London, IET publisher (Renewable series-1), UK, 2010.
2. Magdi S. Mahmoud, Fouad M. AL-Sunni, “Control and Optimization of Distributed Generation Systems”, Springer International Publishing, Switzerland, 2015.
3. Ali K., M.N. Marwali, Min Dai, “Integration of Green and Renewable Energy in Electric Power Systems”, Wiley and sons, New Jersey, 2010.
4. H. Lee Willis, and Walter G. Scott “ Distributed Power Generation: Planning and Evaluation (Power Engineering (Willis) Book 10) , ISBN:978-1482263190, 1st edition, CRC Press,2000
5. Anne-Marie Borbely, and Jan F. Kreider, “Distributed Generation”, ISBN 0-8493-0074-6, CRC Press, 2001,
6. https://nptel.ac.in/courses/108108034

2303EE025	FLEXIBLE AC TRANSMISSION SYSTEMS	L	T	P	C
		3	0	0	3

PREREQUISITE:

Power Electronics and Power Systems

COURSE OBJECTIVES:

- Introduces the application of a variety of high power-electronic controllers for active and reactive power in transmission lines.
- Exposed to the basics, modeling aspects, control and scope for different types of FACTS controllers.

COURSE OUTCOME:

CO1	Explain the concepts of the start-of-art of the power system.
CO2	Describe the performance of VAR compensators and its applications.
CO3	Explain the knowledge on modeling of thyristor controlled series compensators and power flow analysis.
CO4	Understand the performance of Different FACTS Controller and Load flow analysis.
CO5	Summarize the operation of various configurations of Advanced FACTS Controllers.

COs Vs POs MAPPING:

COs	PO1	PO2	PO3	PO4	PO5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11
CO1	3	2	2	2	-	-	1	-	-	-	-
CO2	3	3	3	-	-	-	2	-	-	-	-
CO3	3	3	3	2	-	-	2	-	-	-	-
CO4	3	3	3	2	-	-	2	-	-	-	-
CO5	3	3	3	-	-	-	2	-	-	-	-

COs Vs PSOs MAPPING:

COs	PSO1	PSO2
CO1	2	-
CO2	2	-
CO3	2	-
CO4	2	-
CO5	2	-

MODULE I	**INTRODUCTION**	**9 Hours**

Real and reactive power control in electrical power transmission lines– loads & system compensation-Uncompensated transmission line–shunt and series compensation.

MODULE II	**STATIC VAR COMPENSATOR (SVC) AND APPLICATIONS**	**9 Hours**

Voltage control by SVC–Advantages of slope in dynamic characteristics–Influence of SVC on system voltage–Design of SVC voltage regulator– TCR-FC-TCR- Modelling of SVC for power flow and fast transient stability– Applications: Enhancement of transient stability – Steady state power transfer – Enhancement of power system damping

MODULE III	**THYRISTOR CONTROLLED SERIES CAPACITOR (TCSC) AND APPLICATIONS**	**9 Hours**

Operation of the TCSC–Different modes of operation–Modeling of TCSC, Variability reactance model – Modeling for Power Flow and stability studies. Applications: Improvement of the system stability limit–Enhancement of system damping.

MODULE IV	**VOLTAGE SOURCE CONVERTER BASED FACTS CONTROLLERS**	**9 Hours**

Static Synchronous Compensator (STATCOM)–Principle of operation– V-I Characteristics. Applications: Steady state power transfer-enhancement of transient stability-prevention of voltage instability. SSSC-operation of SSSC and the control of power flow–modeling of SSSC in load flow and transient stability studies - Dynamic voltage restorer (DVR)		
MODULE V	**ADVANCED FACTS CONTROLLERS**	**9 Hours**
Interline DVR (IDVR) - Unified Power flow controller (UPFC) - Interline power flow controller (IPFC) - Unified Power quality conditioner (UPQC).		
FURTHER READING / CONTENT BEYOND SYLLABUS / SEMINAR :		
	Equivalent circuits for FACTS controllers	
REFERENCES		
1. R. Mohan Mathur, Rajiv K Varma, “Thyristor based FACTS Controller for Electrical Power Systems” , John Wiley Sons,2011.		
2.N.G. Hingorani and L. Gyugui “Understanding FACTS Concepts and Technology of Flexible AC Transmission Systems”, B.S. Publications, Indian Reprint 2000		
3.K R Padiyar, “FACTS Controllers in Power Transmission and Distribution”, New Age International Publishers, 2007		
4.J Arriliga and N R Watson, "Computer modeling of Electrical Power Systems”, Wiley, 2001 T J E Miller, "Reactive Power Control in Power Systems'', John Wiley, 1982		

2302EE651	**POWER SYSTEMS LABORATORY**	**L**	**T**	**P**	**C**
		0	**0**	**2**	**1**
PREREQUISITE:					
1	Power system analysis				
2	Power system operation and control				
3	Power system protection and switchgear				
COURSE OBJECTIVES:					
1	Provide hands-on experience with the components of power systems, such as generators, transformers, transmission lines, and circuit breakers.				
2	To simulate and analyze power systems under different operating conditions using tools like MATLAB, PSCAD, or ETAP.				
3	To perform load flow studies, short-circuit analysis, and stability analysis to ensure the reliable operation of power systems.				
COURSE OUTCOMES:					
On the successful completion of the course, students will be able to					
CO1	Analyze the characteristics of IDMT relays, microprocessor-based overvoltage and under voltage relays to Improve the ability of detecting the fault conditions.				
CO2	Analyse the efficiency and performance of transmission lines and calculate ABCD parameters for short, medium, and long lines.				
CO3	Formulate Y_BUS and Z_BUS matrices using software and apply the matrices for load flow and fault studies.				
CO4	Evaluate the AC load flow analysis using the Gauss-Seidel method and Newton-Raphson method and interpret results of voltages and power flows and optimize power system operating conditions.				
CO5	Analyze and simulate different fault conditions in synchronous machines, develop and simulate load frequency control models in MATLAB/Simulink.				
CO6	Develop economic dispatch analysis using software tools and solve optimize fuel costs in power systems. Evaluate DC load flow problems using software under simplified assumptions.				

COs Vs POs MAPPING:

COs	PO1	PO2	PO3	PO4	PO5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11
CO1	3	2	2	2	3	-	-	-	-	-	2
CO2	3	2	2	2	3	-	-	-	-	-	2
CO3	3	3	2	3	3	-	-	-	-	-	2
CO4	3	3	2	3	3	-	-	-	-	-	1
CO5	3	3	3	3	3	-	-	-	-	-	1
CO6	3	2	3	3	3	-	-	-	-	-	2

COs Vs PSOs MAPPING:

COs	PSO1	PSO2
CO1	3	-
CO2	3	-
CO3	3	-
CO4	2	-
CO5	2	-
CO6	3	-

LIST OF EXPERIMENTS:

1	Characteristics of IDMT Over Current Relay.
2	Differential protection of 1-Φ transformer.
3	Characteristics of Micro Processor based Over Voltage/Under Voltage relay.
4	Testing of CT, PT's and Insulator strings.
5	Analyze the performance of a transmission line and compute ABCD parameters.
6	Formation of Y_BUS and Z_BUS matrices using software
7	Formation of AC Load Flow Analysis by Gauss Seidal (GS) method using software.
8	Formation of AC Load Flow Analysis by Newton Raphson method using software.
9	Single Line-Ground, Line-Line and 3-Φ fault analysis of 3-Φ synchronous machine.
10	Develop the SIMULINK model and simulate the single and two area load frequency control problem using software.
11	Economic Dispatch in Power Systems using software
12	Study on DC load flow problem using software.

TOTAL: 30 HOURS

REFERENCES:

1. HadiSaadat, Power System Analysis, 2/e, McGraw Hill, 2002.
2. M. S. Naidu, V. Kamaraju, High Voltage Engineering, Tata McGraw-Hill Education, 2004.
3. Wadhwa C. L., Electrical Power Systems, 3/e, New Age International, 2009.
4. Kothari D. P. and I. J. Nagrath, Modern Power System Analysis, 2/e, TMH, 2009
5. https://archive.nptel.ac.in/courses/108/105/108105104/
6. https://vp-dei.vlabs.ac.in/Dreamweaver/list.html

2304EE652	MINI PROJECT	L	T	P	C
		0	0	2	1

CO1: Identify and define a Real-world Electrical and Electronics Engineering problem through literature survey.
CO2: Design and model an electrical/electronic system using appropriate software tools.
CO3: Analyze, simulate and validate the performance of the proposed system.
CO4: Develop and test a prototype/system considering safety, environmental and ethical aspects.
CO5: Prepare technical documentation and demonstrate effective presentation skills.

COs Vs. POs MAPPING:

COs	PO1	PO2	PO3	PO4	PO5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11
CO1	3	2	1	1	-	-	-	-	-	-	-
CO2	2	3	3	2	3	-	-	-	-	-	-
CO3	3	3	2	3	2	-	-	-	-	-	-
CO4	2	2	3	2	2	2	2	2	2	-	1
CO5	-	-	-	-	-	-	-	2	3	3	1

COs Vs PSOs MAPPING:

COs	PSO1	PSO2
CO1	3	3
CO2	3	3
CO3	3	3
CO4	3	3
CO5	-	-

GUIDELINE FOR REVIEW AND EVALUATION

The students may be grouped into 2 to 4 and work under a project supervisor. The device / system / component(s) to be designed and developed using modeling software, may be decided in consultation with the supervisor and if possible, with an industry. A project report to be submitted by the group and the soft copy of the report, which will be reviewed and evaluated for internal assessment by a committee constituted by the Head of the Department. Two internal reviews and one External review with oral viva & project report will be examined.

TOTAL: 30 Hours

2301MC613	**MENTAL HEALTH AND EMOTIONAL WELL-BEING**	**L**	**T**	**P**	**C**
		2	**0**	**0**	**0**

COURSE OBJECTIVES:

1.	Understand key concepts and factors influencing mental health and wellbeing.
2.	Learn practical strategies and interventions for promoting holistic wellbeing.

COURSE OUTCOMES:

Upon successful completion of the course, students will be able to

CO1	Explain the foundations of mental health, its key influences, and factors that build resilience.
CO2	Demonstrate effective coping strategies, communication skills, and techniques for managing stress.
CO3	Apply strategies for promoting wellbeing, providing crisis support, and maintaining digital balance.

COURSE CONTENTS:

Module-I	**FOUNDATIONS OF MENTAL HEALTH AND EMOTIONAL WELL-BEING**	**9 Hours**

Adolescent mental health – Identity formation – Social and academic pressures – Role of relationships and support systems – Stress management techniques – Impact of technology and social media – Self-care practices – Self-compassion – Positive self-talk.

Module-II	**COPING MECHANISMS, LIFE SKILLS AND RESILIENCE**	**9 Hours**

Healthy coping strategies – Emotional regulation – Peer influence – Body image and self-esteem – Mindfulness – Communication skills – Conflict resolution – Resilience in academic transitions – Time management – Sleep, nutrition and exercise for mental well-being.

Module-III	**ADVANCED MENTAL HEALTH SKILLS AND SUPPORT SYSTEMS**	**9 Hours**

Academic and career anxiety – Therapy and counseling – Emotional intelligence in leadership – Boundary-setting – Emotional validation – Crisis support – Suicide prevention – Fostering resilience – Burnout prevention – Crisis management skills.

Module-IV	**CREATIVE EXPRESSION, RELATIONSHIPS AND ATTACHMENT**	**9 Hours**

Creative arts for well-being – Journaling and storytelling – Movement-based therapies – Managing exam anxiety – Attachment styles and relationships – Work-life balance – Building healthy friendships – Healing from relationship trauma – Parenting and intergenerational patterns.

Module-V	**STRESS MANAGEMENT, DIGITAL WELL-BEING AND MOTIVATION**	**9 Hours**

Motivational enhancement – Academic and professional relationship-building – Cultural sensitivity – Impact of social media on self-esteem and body image – Mental health risks from digital overuse – Healthy digital habits – Screen time boundaries – Positive online engagement.

TOTAL: 45 HOURS

REFERENCES:

1. Bessel van der Kolk, "The Body Keeps the Score: Brain, Mind, and Body in the Healing of Trauma," ISBN: 0143127748, 2015
2. Ronald J. Comer, "Abnormal Psychology," ISBN: 1319190723, 2021
3. Kathleen O'Hara, "Emotion Regulation: Conceptual and Practical Issues," ISBN: 0198709633, 2020
4. Cathy A. Malchiodi (Ed.), "Creative Arts and Play Therapy for Attachment Problems," ISBN: 1462523702, 2013
5. Thomas Joiner, "Why People Die by Suicide," ISBN: 0674061981, 2005

OPEN ELECTIVE

2303EE031	ELECTRONIC WASTE MANAGEMENT ISSUES AND CHALLENGES	L	T	P	C
		3	0	0	3

PREREQUISITE

Environmental Science

COURSE OBJECTIVE

1	To understand the overview of electronic waste
2	To understand the impact and management of E waste

COURSE OUTCOME

CO1	Explain the evolution of E-waste and presence of hazardous substances in electrical And electronic equipment.
CO2	Analyze the toxicity characteristics of E-waste and evaluate its effects on human health, public safety.
CO3	Examine the E-waste management practices, regulatory frameworks, and risk assessment strategies
CO4	Compare various E-waste recycling and material recovery technologies
CO5	Evaluate national and international E-waste legislation and assess their Effectiveness in achieving sustainable E-waste management.

COs Vs POs MAPPING:

COs	PO1	PO2	PO3	PO4	PO5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11
CO1	3	3	-	-	-	2	3	2	-	-	-
CO2	3	3	-	-	-	2	3	2	2	-	-
CO3	2	3	-	-	-	3	3	-	-	-	-
CO4	2	3	-	-	-	3	3	-	-	1	-
CO5	2	3	-	-	-	3	2	2	-	1	-

COs Vs PSOs MAPPING:

COs	PSO1	PSO2
CO1	-	3
CO2	-	2
CO3	-	2
CO4	-	2
CO5	-	1

MODULE I	**OVERVIEW OF ELECTRONIC WASTE**	**9 Hours**

E-waste growth; Digital dump yard; Minimization of E-waste; Hazardous substances waste- Electrical and electronic equipment, batteries, plastic and flame retardants, circuit boards; Characteristics of pollutants, Pollutants in waste electrical and electronic equipment.

MODULE II	**ENVIRONMENTAL AND PUBLIC HEALTH ISSUES**	**9 Hours**

WEEE-flows; Quantity; Characteristics of a WEEE; Socio economic matters; Indian and international perspective; Health and safety implications; Toxicity concerns; Hazardous substances; E-waste health risk assessment; Case study.

MODULE III	**E-WASTE MANAGEMENT IN INDIA**	**9 Hours**

Current Indian scenario; Environmental regulations for E-waste in India; Classification of E-waste; Components of E-waste; E-waste recycling and technology currently used in India; Mechanical processing and Biotechnology; Awareness creation; Challenges faced by formal recyclers in Delhi-NCR; Case study.

MODULE IV	**RECYCLING AND RECOVERY OF MATERIALS FROM E-WASTE**	**9 Hours**
Recycling process for the recovery of metal and materials; Recovery and recycling technologies; Bioleaching and biotechnological initiatives; Nano particles synthesis; Hydrometallurgical techniques; Pyrometallurgy.		
MODULE V	**BEST PRATICES IN EFFICIENT MANAGEMENT OF E-WASTE**	**9 Hours**
Current practices of E-waste management in different countries- China, Brazil, Argentina, Nigeria, Pakistan, Srilanka; Policy comparison between developing and developed countries; Sustainable E-waste management.		
	TOTAL:	**45 HOURS**

REFERENCES:

1. Rakesh Johri, “E-waste: Implications, Regulations and Management in India and Current Global Best Practices”, The Energy and Resources Institute (TERI) press, New Delhi, 2008.
2. M.N.Vara Prasad and Meththika Vithanage, “Electronic Waste Management and Treatment Technology”, Butterworth-Heinmann (Elsevier), 2019.
3. R.E.Hester and R.M.Harrison, “ Electronic Waste Management”, RSC publishing, 2009
4. MeththikaVithanage and Anwesha Borthakur, “Handbook of Electronic Waste Management”, Elsevier Science, 2019.
5. Inamuddin and Abdullah M. Asiri, “E-waste Recycling and Management”, Springer International Publishing, 2019.
6. https://nptel.ac.in/courses/105/105/105105169/